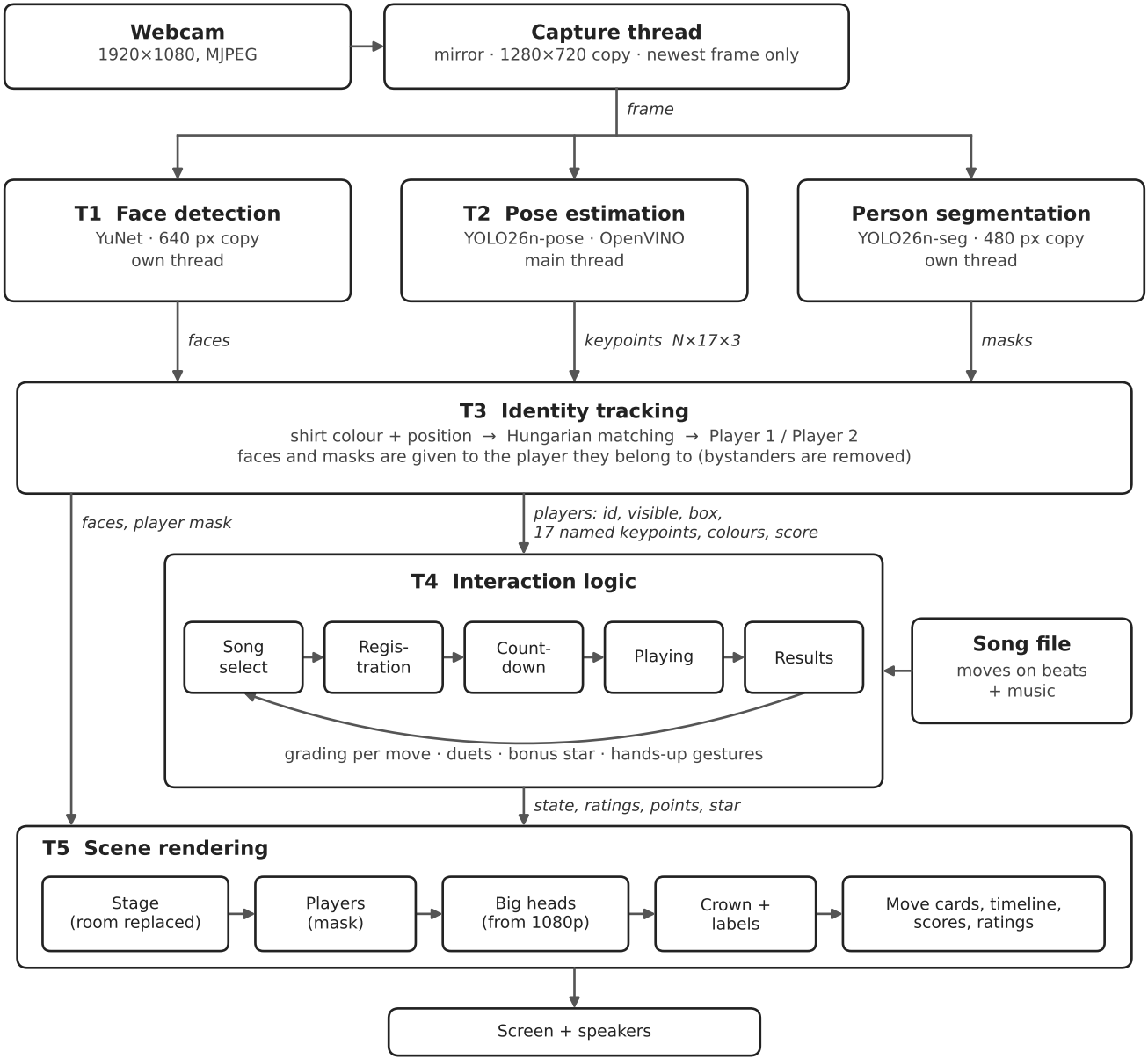
Webcam
1920×1080, MJPEG
Capture thread
mirror · 1280×720 copy · newest frame only
frame
T1 Face detection
YuNet · 640 px copy
own thread
T2 Pose estimation
YOLO26n-pose · OpenVINO
main thread
Person segmentation
YOLO26n-seg · 480 px copy
own thread
faces
keypoints N×17×3
masks
T3 Identity tracking
shirt colour + position → Hungarian matching → Player 1 / Player 2
faces and masks are given to the player they belong to (bystanders are removed)
faces, player mask
players: id, visible, box,
17 named keypoints, colours, score
T4 Interaction logic
Song select
Regis-tration
Count-down
Playing
Results
Song file
moves on beats
+ music
grading per move · duets · bonus star · hands-up gestures
state, ratings, points, star
T5 Scene rendering
Stage
(room replaced)
Players
(mask)
Big heads
(from 1080p)
Crown +
labels
Move cards, timeline,
scores, ratings
Screen + speakers